

How we use chart types Transcript

Now, as users of technology and users of special software for doing visualisations, we will be now moving to the part where we are taking use of this algorithms or techniques or softwares and build the visualisation on the textual data. So, the example that we will be doing today in class as a theory session is the Panchatantra story text analysis. Same or similar exercise will be done in your experiential hands on session, so that you get a feel of how to process the text data and convert and draw visualisations from the text data.

So, in Panchatantra story, you have the chapters 5 chapters or the 5 types of stories which are different types of strategies Mitra Veda, Mitra Laba, so all there are 5 types of stories, each one of them looking at a particular strategies separation of friends, gain of friends, war and peace, loss of gains and ill consider actions. So, these are different types of strategies that are there and each has a story or a storyline collection of stories, then you have the full text, we have the full text of the story and this is how the data is organised. So, for us at the moment, we are not looking into the characterisation of the text and how is it converted into the compute friendly form, we are now just simply applying particular algorithms and trying to understand the data.

We will be treating the text as a underlying data set and then try to do all our data visualisation function, so that we can portray the information that is there in the data set in a condesceive format to our audience. So, first thing is doing a word cloud, word cloud is the most common form of visualising text data, it simply shows the dominant word in a larger font or in a highlighted font. For example, once you run the word cloud on the panchatantra text, you would see that the words like king, friend, lion, one, jackal, wife, these are the words that are seen quite often in the text and the word king is seen in the is the largest word that you can immediately see or perceive when you see the word cloud.

Correspondingly, you also see on onto your right the bar chart, which shows the number of times the word is used, for example, the word king is used 250 times as compared to friend which is used only 100 times, then you have day, jackal, lion, so these are the difference you can understand that. Word cloud represents the word that is used or the main character in this case, which is the main character in the panchatantra text, you have the word cloud representing the main characters king, friend, lion are the main characters depending upon how often they have been referred or used. Similarly, you have the number of most frequently used nouns, though this is not a one to one correlation between these two charts, just want to show that the king being a proper noun is used 250 times is also the dominant word in the or dominant character in the panchatantra.



Likewise, you have a friend, which is dominant in terms of a main character, but it is not a noun that is used most frequently. So, the construction differs a bit between these two graphs. However, these are different ways in which you can understand what is the underlying data or text.

So, the main character, so somebody gave you a panchatantra book and then you wanted to understand what are the main characters, you can simply run a word cloud and see which are the main characters or they asked you what are the different types of character names or nouns that you have. So, you can simply run a most frequent used noun and count them, then you get this kind of a distribution and can simply draw a conclusion that the word king has been used multiple times. So, we now move to the different types of visualising text data, again bar chart.

This time we will try to see in each of the text what are the different type of number of words. First text Miturveda has 13000 almost 14000 words, whereas Mitralabha has only 8300 words. The length of the text and the complexity that is there, the 5 texts are there, 5 stories are there which is having more words.

And then we come to the word tree which is simply trying to see how different characters are related to each other through text. Lion king and Dhamanaka, lion king directly relates to Dhamanaka and Dhamanaka relates further to some other characters. This is done through arranging the text and identifying the characters relationship within that.

Now, we are not getting into the process how is it done, but this is how we are showing the lion king and Dhamanaka are related closely as opposed to lion king and somebody else in the third sentence. So, this is how a word tree propagates, you start with a node, then you see which is the closest and what are the relations. So, it is like it is branching out, you can see the relationship moving across the text.

This is the same way you would feel when you read a text, you would see one character related to the other character, but the direction relationship might not be immediately visible in the text format, but whereas a word tree format tries to place them closest. Now, we have semantics course which is a very important part of textual analysis. Whenever we say a text, it either has a positive tone, neutral tone or a negative tone.

Administrative nodes are neutral in tone, whereas emotional messages are either positive or negative. When you have a complaint, you expect the customer to have negative tone, when you have a good rating, you expect the customer to give positive feedback. So, whenever there is a text, there is something called tonality or the sentiment.

So, the algorithms can give sentiment scores depending upon the tone of the sentence and aside what is the overall sentiment of the text, whether the entire text is being positive or negative or neutral. In this case, each of these chapters or the stories have been given a sentiment score and on an average if you see, if you put all the scores together, the Panchatantra seems to be a neutral tone story, because all the scores get cancelled out, you have almost neutral story. But within the segments, for example, Mithralabha, the gaining of friends is expected to have a positive sentiment, is having a positive sentiment as opposed to this Mithrabheda separation of friends, sadness and repent, that can have a negative tone.

This is clearly reflected by the sentiment scores. One another example that often used in the sentiment score space is the analysis of Twitter feeds or the customer feedback review or for that matter when you buy anything on Amazon, there is a simple summary of the feedbacks and reviews and the review ratings and there is a final comments that most of the customers feel good about this product, most of the customers feel the product is not worth to the money. Such statements are based on both mixed analysis of sentiment scores and the text analytics that the reviews are consolidated and you get a final output from that.

So, this is the use of sentiment scores and it is a very interesting space for improving customer feedback. Now we will come to something called use of joint plots. Joint plots are like scatter diagrams, however, they are a bit different.

They also have the distribution of the variables X and Y on the other axis. Now in this joint plot, we will be under studying or plotting the polarity and subjectivity. Polarity and subjectivity are again like sentiment scores in a different form.

Polarity also ranges from minus 1 to plus 1 like the sentiment score, whereas subjectivity ranges from 0 to 1. Subjectivity tries to see whether the sentence is based on objective things or it has more of emotions and feelings. Greater is the emotion or feeling in the sentence, it would push the score towards one saying that it is subjective in nature. I feel hungry right, it is a subjective statement.

I get some salary that is an objective statement therefore, I feel rich. So, these are objective statements, the subjectivity score there would be less. So, this way the algorithm would give subjectivity scores to the text that is supplied to it.

In this case, we have given the Panchatantra text to the algorithm and you can see the distribution of the scores shows that is generally as the polarity is increasing, the subjectivity is also increasing. So, when you become too strong or too weak, too positive or too negative, the subjectivity of your statements is also increasing and on an average, the polarity statements are distributed almost normally.



However, the subjectivity statements are not distributed normally, there is some amount of skewness or looks like a multimodal distribution.

So, the benefit of using a joint plot in a text is you can understand the relationship between either subjectivity or polarity and simultaneously understand the underlying distribution. Now, we come to another important topic within the textual visualising of text data, how are conversations done? Whenever you talk about a conversation, there are five aspects when you discuss any conversation. First part is when did the conversation happen and who are the parties to the conversation, then from whom to whom the conversation has gone, at what point in time, then what is the density of that conversation like how many people are there, whether there is multifold communication between the parties who are involved in that conversation, whether it is between two people, whether it is two to three people or the same person is talking to 5-6 people.

So, this is a network of communication that is happening and we would like to understand as a conversation how is it flowing through that particular network or what are the parties who are involved in that network and finally, you are interested in what is being spoken about or what is the content of the conversation. So, again whenever we are looking at a conversation, there are five aspects we are interested in. First of all, when is it when was it happened or when is it happening, whenever we are looking at conversations, we are looking at five aspects.

First is when did the conversation happen, who are the people in the conversation, when did they speak with each other and who and who have responded or who and who are involved in this conversation and finally, what did they speak about. So, these five characteristics capture the essence of any conversation and we will see how you can use visualisation tools on text data to show these features in a conversation. First and foremost, we take an example from the Enron emails data set, Enron is a American company which has declared bankruptcy and there were lot of discussions prior to this bankruptcy among the senior staff within the company, they were then taken into scrutiny by the US regulator and found malfeasance and they have been fined.

So, the story apart the Enron email story clearly shows that you can while analysing the text data, you can clearly identify patterns about the company and its functionality. So, the same thing can be extended to understanding the customer reviews or understanding between discussions between a network of friends or within an organisation using emails. The content of the text is not relevant, but how are we going to solve this problem or what are the tools that are available for us to present this are very interesting.



So, in this case, first we will start to understand the question when did the conversation happen. Enron declared bankruptcy in late 2001 and before 2002, somewhere around this period is when Enron has declared bankruptcy. So, you can see the increase in the email communication in the Enron's account just before the bankruptcy clearly showing a trend change.

So, for somebody who is looking at this irrespective of what is being spoken can understand some other phenomenon is brewing or some insight is there which we need further information to understand. So, this is a simple visualisation of text data by counting the number of emails, this is like counting the number of documents, then presenting it on a timeline. So, you have used a simple bar chart and shown how many emails are there across the dates, then who which people have sent these emails.

So, there are the top management people. So, Derek is the CEO, he has sent the maximum number of emails about 43 percent and you have the others who are the share of our email sent by other top management people in the company. Then you are trying to understand who and when how many emails has Derek sent per month.

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